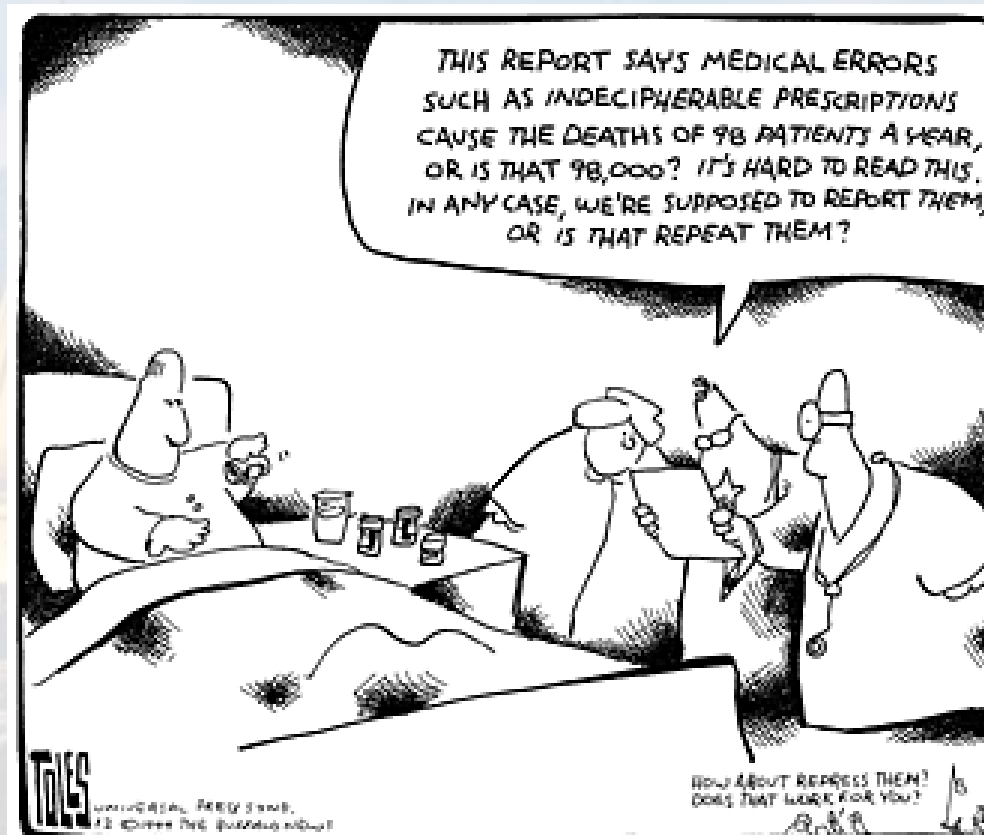


M. Hohle:

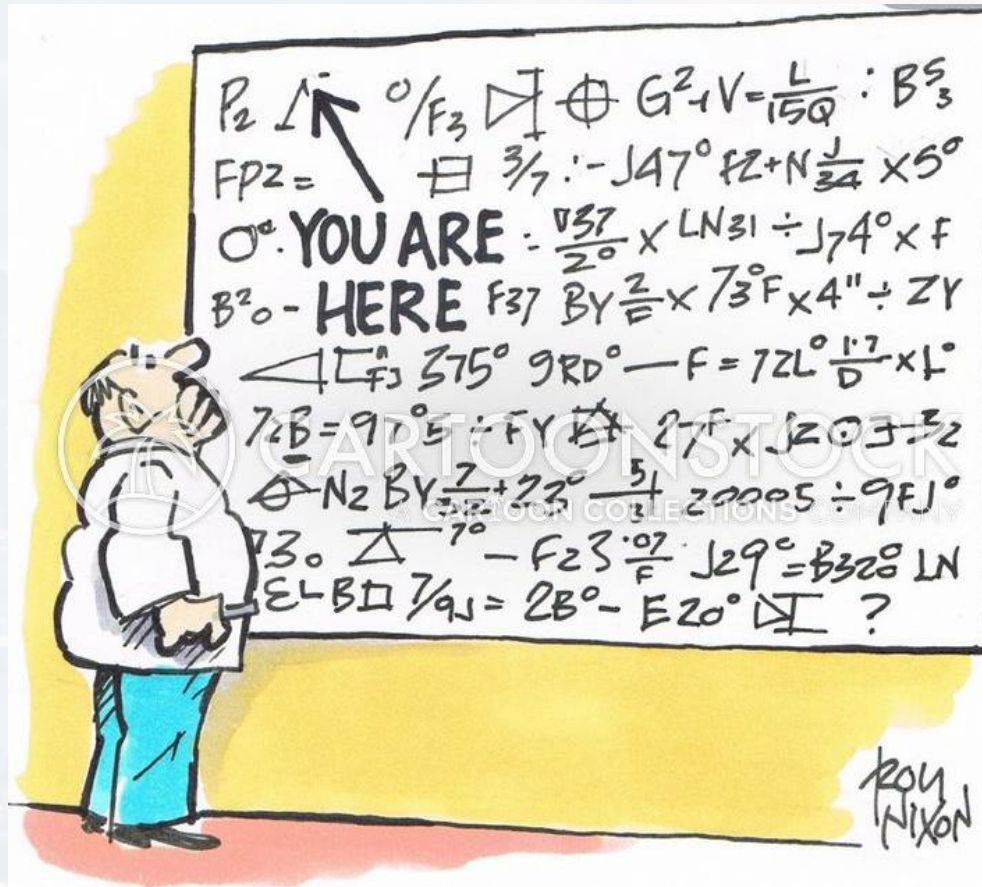
Physics 77/88: Introduction to Computational Techniques in Physics



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Week	Date	Topic
1	Aug 26 th	Programming Environment & UIs for Python. Programming Fundamentals
2	Sep 2 nd	Basic Types in Python
3	Sep 9 th	Parsing, Data Processing and File I/O/ Visualization: Matplotlib & Seaborn
4	Sep 16 th	Functions, Map & Lambda
5	Sep 23 rd	Scientific Computing with numpy and avoiding Loops
6	Sep 30 th	Random Numbers & Probability Distributions
7	Oct 7 th	Interpreting Measurements & Introduction to Bayesian Statistics
8	Oct 14th	Numerical Integration and Differentiation
9	Oct 21 st	Root finding, Interpolation
10	Oct 28 th	Systems of Linear Equations
11	Nov 4 th	Introduction to Ordinary Differential Equations (ODEs)
	Nov 11 th	<i>Veteran's Day</i>
12	Nov 18 th	Stability of ODEs, Runge-Kutta, Examples
	Nov 25 th	<i>Thanksgiving</i>
13	Dec 2 nd	Final Project Presentations



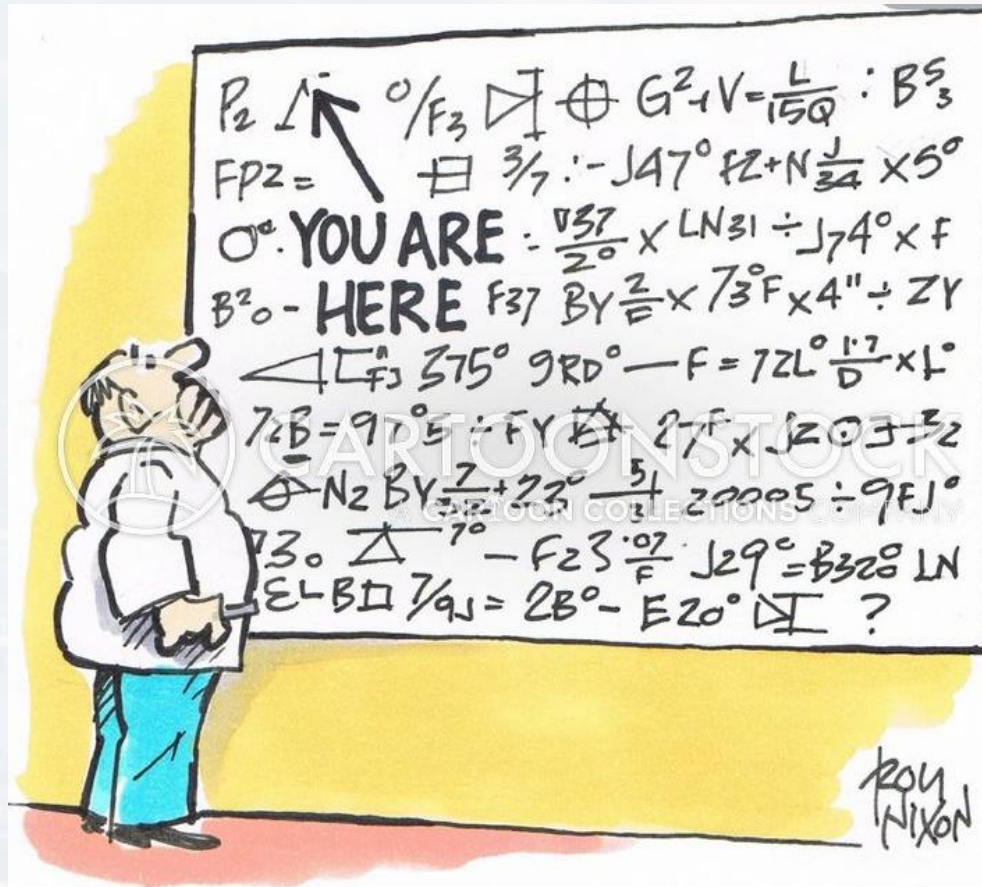
Outline:

Derivatives

Taylor Series

Integrals

Diffusion Example



Outline:

Derivatives

Taylor Series

Integrals

Diffusion Example



motivation:

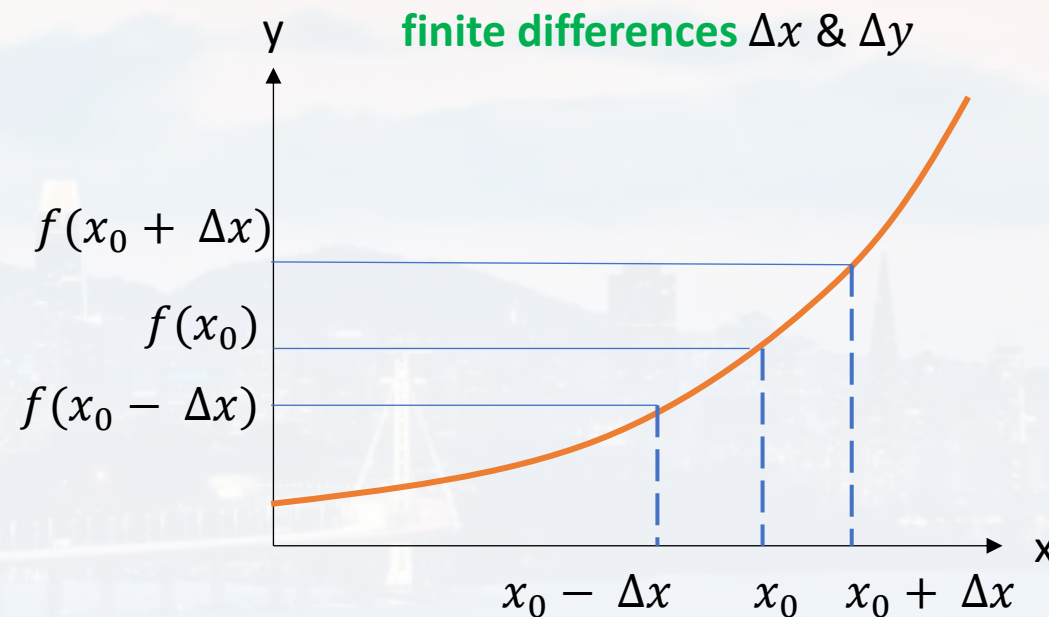
- optimization algorithms (gradient descent and related)
- ANNs learn via **backpropagation** → chain rule
- approximation methods (Taylor series)
- maximum entropy distributions (data analysis, data modelling)
- error estimation and error propagation
- and more...



slope of a function at $x = x_0$

$$\left. \frac{df^+}{dx} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

$$\left. \frac{df^-}{dx} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0) - f(x_0 - \Delta x)}{\Delta x}$$



$$\left. \frac{df}{dx} \right|_{x=x_0} = \frac{1}{2} \left(\left. \frac{df^+}{dx} \right|_{x=x_0} + \left. \frac{df^-}{dx} \right|_{x=x_0} \right) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0 - \Delta x)}{2\Delta x}$$

1st derivative at $x = x_0$

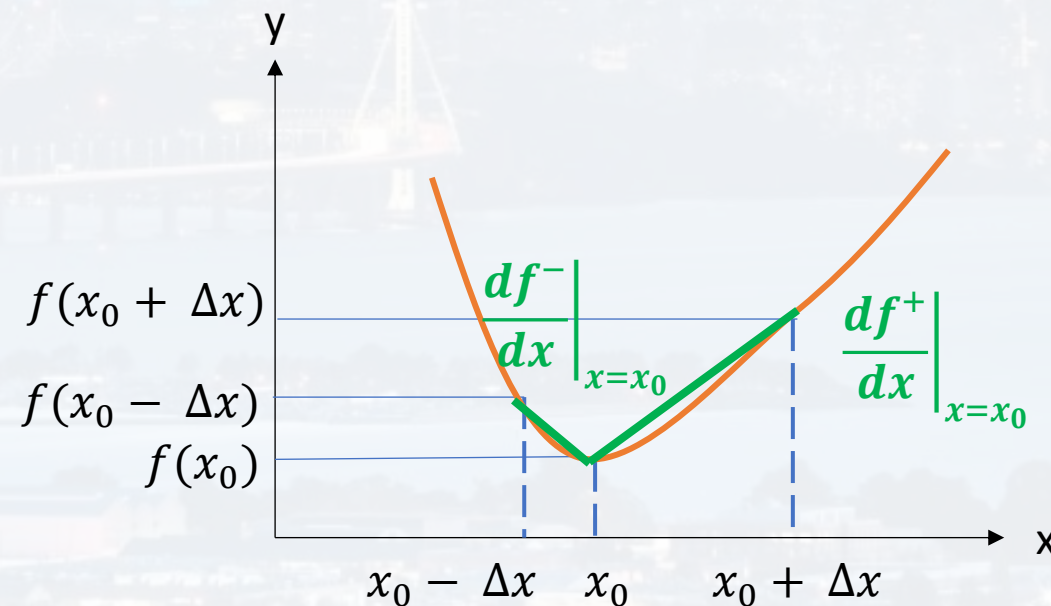


$$\left. \frac{df}{dx} \right|_{x=x_0} = \frac{1}{2} \left(\left. \frac{df^+}{dx} \right|_{x=x_0} + \left. \frac{df^-}{dx} \right|_{x=x_0} \right) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0 - \Delta x)}{2\Delta x}$$

1st derivative at $x = x_0$

change of the slope of a function at $x = x_0$, aka curvature

$$\left. \frac{d^2 f}{dx^2} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{1}{\Delta x} \left(\left. \frac{df^+}{dx} \right|_{x=x_0} - \left. \frac{df^-}{dx} \right|_{x=x_0} \right)$$





$$\left. \frac{df}{dx} \right|_{x=x_0} = \frac{1}{2} \left(\left. \frac{df^+}{dx} \right|_{x=x_0} + \left. \frac{df^-}{dx} \right|_{x=x_0} \right) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0 - \Delta x)}{2\Delta x}$$

1st derivative at $x = x_0$

change of the slope of a function at $x = x_0$, aka *curvature*

$$\left. \frac{d^2 f}{dx^2} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{1}{\Delta x} \left(\left. \frac{df^+}{dx} \right|_{x=x_0} - \left. \frac{df^-}{dx} \right|_{x=x_0} \right) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - 2f(x_0) + f(x_0 - \Delta x)}{\Delta x^2}$$

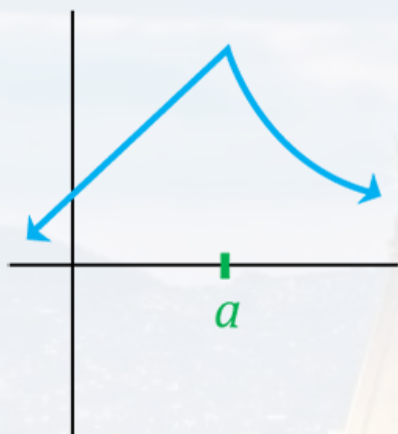
$$\left. \frac{d^2 f}{dx^2} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - 2f(x_0) + f(x_0 - \Delta x)}{\Delta x^2}$$

2nd derivative at $x = x_0$

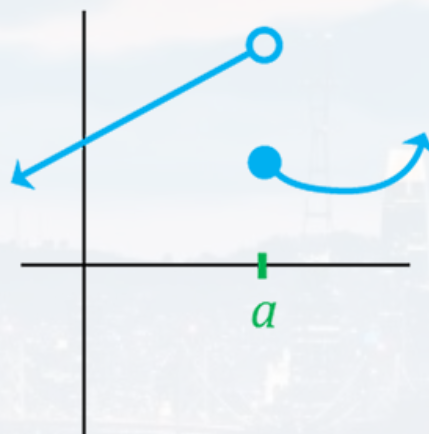
...and so on



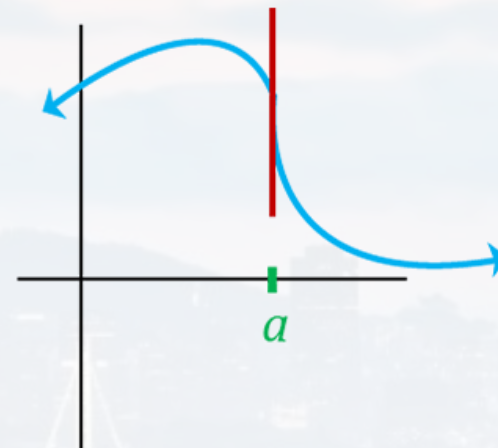
derivatives are not always defined:



Cusp / Corner

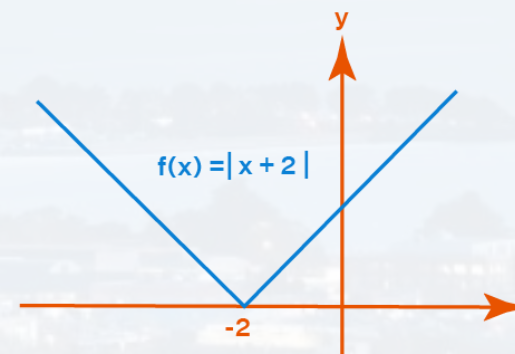


Discontinuous



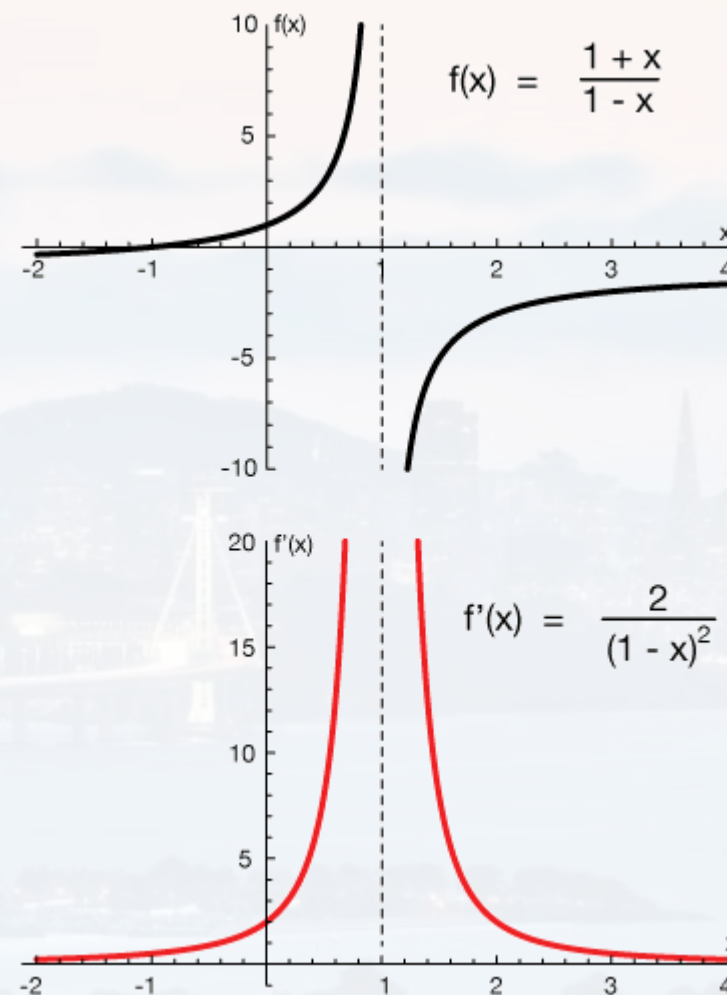
Vertical Tangent

function needs to be **continuous and differentiable**





derivatives are not always defined:



function needs to be **continuous and differentiable**



example: $f(x) = \sqrt{x}$

$$\begin{aligned}\left. \frac{df}{dx} \right|_{x=x_0} &= \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0 - \Delta x)}{2\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\sqrt{x_0 + \Delta x} - \sqrt{x_0 - \Delta x}}{2\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{(\sqrt{x_0 + \Delta x} - \sqrt{x_0 - \Delta x})(\sqrt{x_0 + \Delta x} + \sqrt{x_0 - \Delta x})}{2\Delta x (\sqrt{x_0 + \Delta x} + \sqrt{x_0 - \Delta x})} \\ &= \lim_{\Delta x \rightarrow 0} \frac{x_0 + \Delta x - x_0 + \Delta x}{2\Delta x (\sqrt{x_0 + \Delta x} + \sqrt{x_0 - \Delta x})} \\ &= \lim_{\Delta x \rightarrow 0} \frac{2\Delta x}{2\Delta x (\sqrt{x_0 + \Delta x} + \sqrt{x_0 - \Delta x})} = \frac{1}{2\sqrt{x_0}}\end{aligned}$$



rules: $\frac{d}{dx} ax^n = a n x^{n-1}$

$$a \in \mathbb{C}$$
$$n \in \mathbb{R}$$

$$\frac{d}{dx} [f(x) + g(x)] = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$$

sum rule: derivatives are linear

$$\frac{d}{dx} [f(x)g(x)] = g(x) \frac{d}{dx} f(x) + f(x) \frac{d}{dx} g(x)$$

product rule

$$\frac{d}{dx} \mathbf{f[g(x)]} = \frac{df(x)}{dg(x)} \frac{d}{dx} g(x)$$

chain rule

outer derivative **inner derivative**



special derivatives

$$a \in \mathbb{C}$$

$$n \in \mathbb{R}$$

$$\frac{d}{dx} e^x = e^x$$

the actual definition of e

$$\frac{d}{dx} b^x = \ln(b) b^x$$

$$b > 0$$

$$\frac{d}{dx} \log_b(x) = \frac{1}{x \ln(b)}$$

$$\frac{d}{dx} \sin(x) = \cos(x)$$

$$\frac{d}{dx} \cos(x) = -\sin(x)$$



$$a \in \mathbb{C}$$

$$n \in \mathbb{R}$$

$$\frac{d}{dx} ax^n = a n x^{n-1}$$

$$\frac{d}{dx} 3x^5 = 15x^4$$

$$\frac{d}{dx} [f(x) + g(x)] = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$$

$$\frac{d}{dx} [3x^5 - 2x] = 15x^4 - 2$$

$$\frac{d}{dx} [f(x)g(x)] = g(x) \frac{d}{dx} f(x) + f(x) \frac{d}{dx} g(x)$$

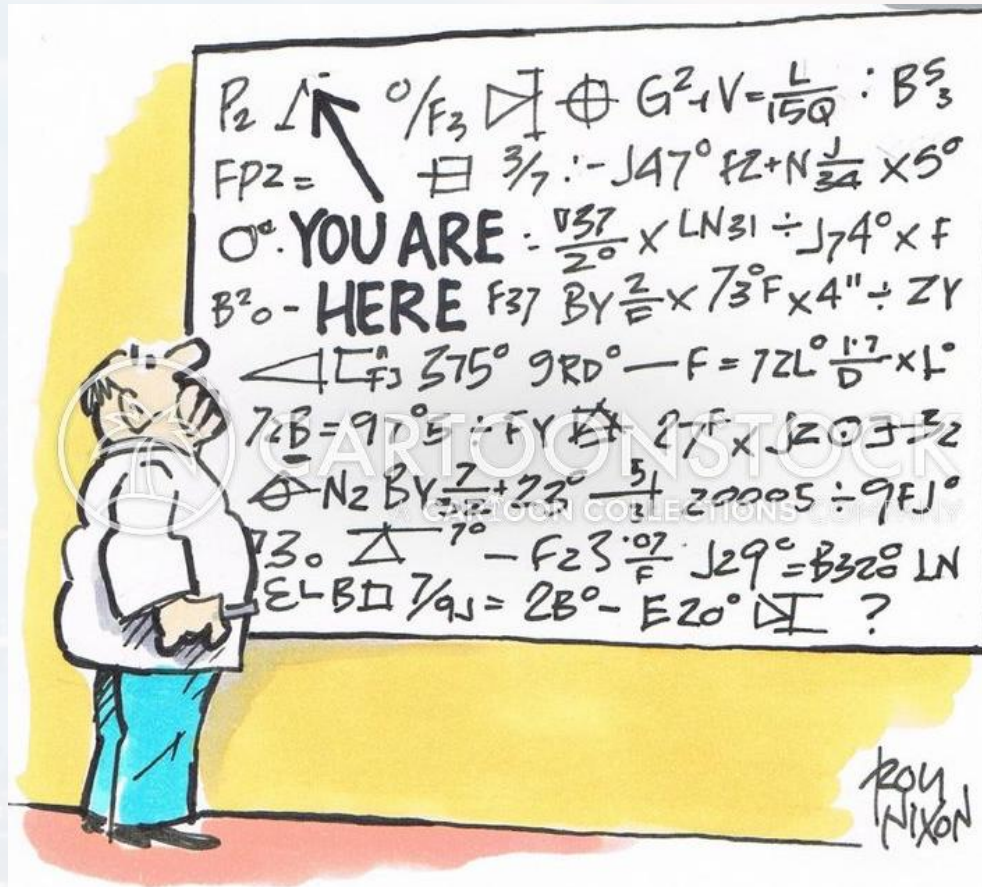
$$\frac{d}{dx} [3x^5 \sin(x)] = 15x^4 \sin(x) + 3x^5 \cos(x)$$

$$\frac{d}{dx} f[g(x)] = \frac{df(x)}{dg(x)} \frac{d}{dx} g(x)$$

outer derivative inner derivative

$$\frac{d}{dx} \sin(3x^5) = \cos(3x^5) 15x^4$$

outer derivative inner derivative



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Diffusion Example



$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n f}{dx^n} \right|_{x=x_0} (x - x_0)^n$$

$$n = 0: \quad f(x) \approx f(x_0)$$

$$n = 1: \quad f(x) \approx f(x_0) + \left. \frac{df}{dx} \right|_{x=x_0} (x - x_0) \quad \text{tangent on } f \text{ at } x = x_0$$

$$\frac{f(x) - f(x_0)}{x - x_0} \approx \left. \frac{df}{dx} \right|_{x=x_0}$$

$$\left. \frac{df}{dx} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

definition of the 1st derivative!



$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n f}{dx^n} \right|_{x=x_0} (x - x_0)^n$$

$$n = 0: \quad f(x) \approx f(x_0)$$

$$n = 1: \quad f(x) \approx f(x_0) + \left. \frac{df}{dx} \right|_{x=x_0} (x - x_0) \quad \text{tangent on } f \text{ at } x = x_0$$

$$n = 2: \quad f(x) \approx f(x_0) + \left. \frac{df}{dx} \right|_{x=x_0} (x - x_0) + \frac{1}{2} \left. \frac{d^2 f}{dx^2} \right|_{x=x_0} (x - x_0)^2$$



$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n f}{dx^n} \right|_{x=x_0} (x - x_0)^n$$

$$n = 0: \quad f(x) \approx f(x_0)$$

$$n = 1: \quad f(x) \approx f(x_0) + \left. \frac{df}{dx} \right|_{x=x_0} (x - x_0) \quad \text{tangent on } f \text{ at } x = x_0$$

$$n = 2: \quad f(x) \approx f(x_0) + \left. \frac{df}{dx} \right|_{x=x_0} (x - x_0) + \frac{1}{2} \left. \frac{d^2 f}{dx^2} \right|_{x=x_0} (x - x_0)^2$$

exercise:

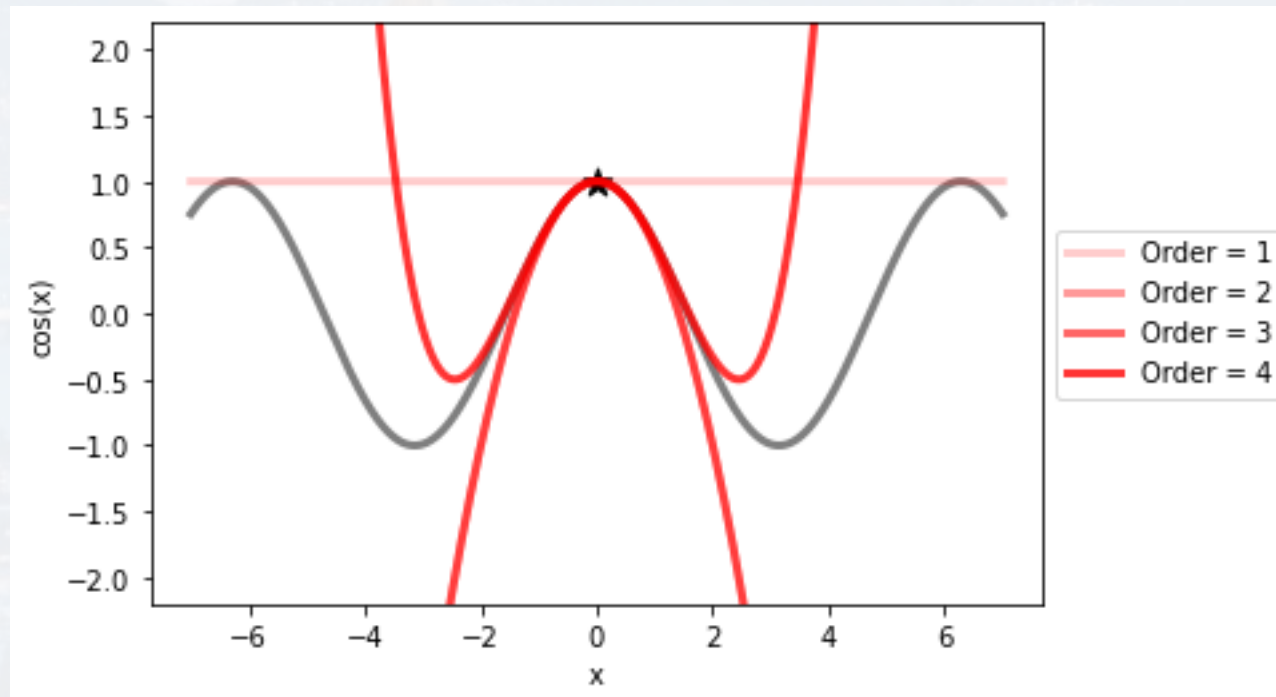


- write down the Taylor Series of $\sin(x)$, $\cos(x)$ and e^x at $x_0 = 0$
- express all three series as an infinite sum
- try to combine all three equations by introducing a new mathematical object i which only property is $i^2 = -1$



$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n f}{dx^n} \right|_{x=x_0} (x - x_0)^n$$

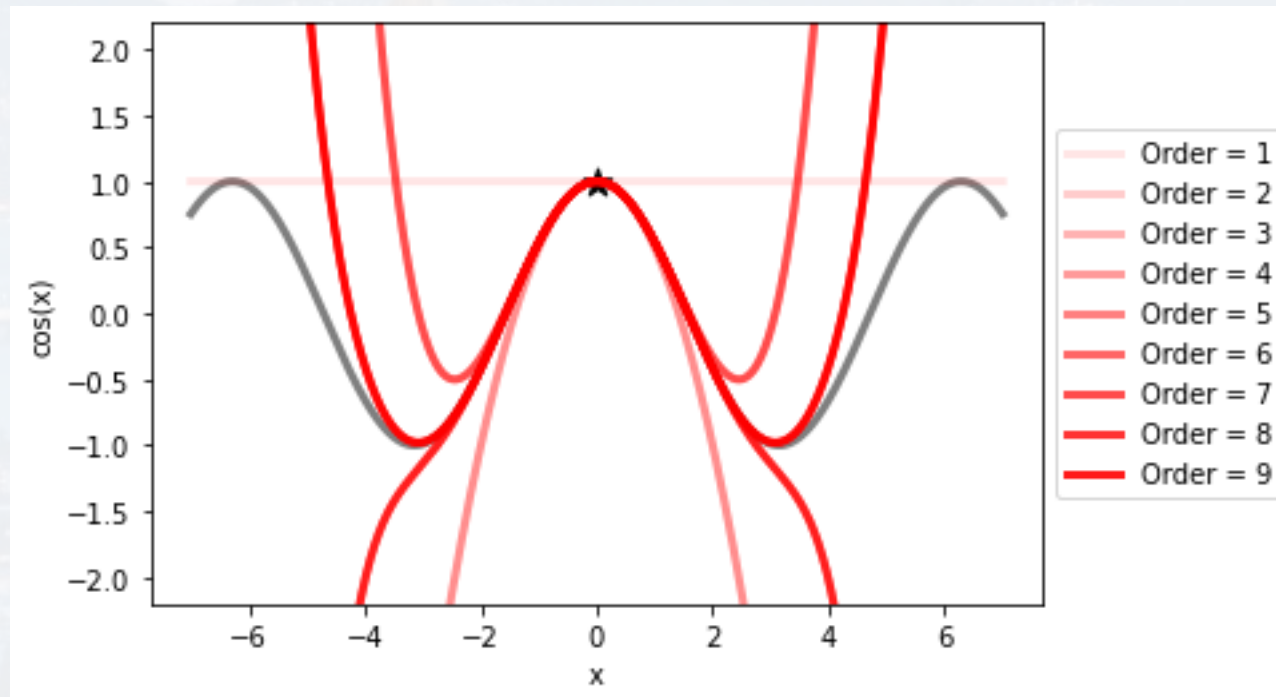
run the function `PlotTaylorSeries.py`





$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n f}{dx^n} \right|_{x=x_0} (x - x_0)^n$$

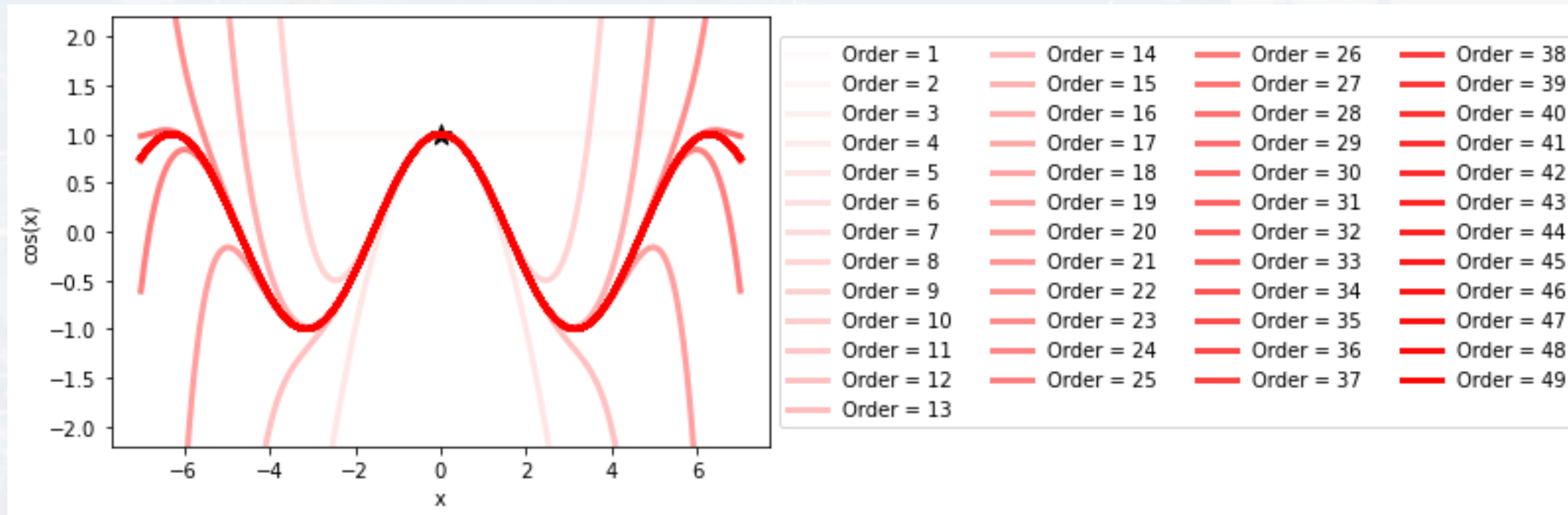
run the function `PlotTaylorSeries.py`

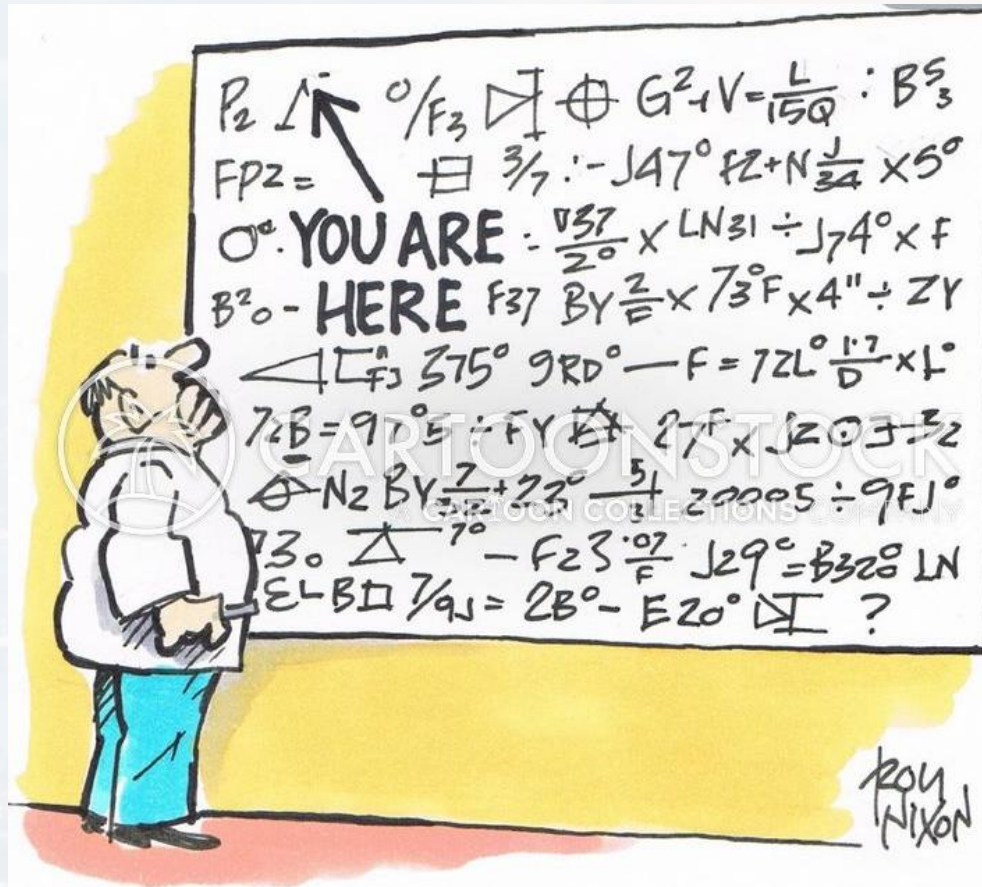




$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n f}{dx^n} \right|_{x=x_0} (x - x_0)^n$$

run the function `PlotTaylorSeries.py`





Outline:

Derivatives

Taylor Series

Integrals

Diffusion Example



motivation:

- deriving probabilities from likelihood functions
- normalization tools
- calculating volumes, areas, flow, energy, etc....
- sums $\rightarrow$ integral

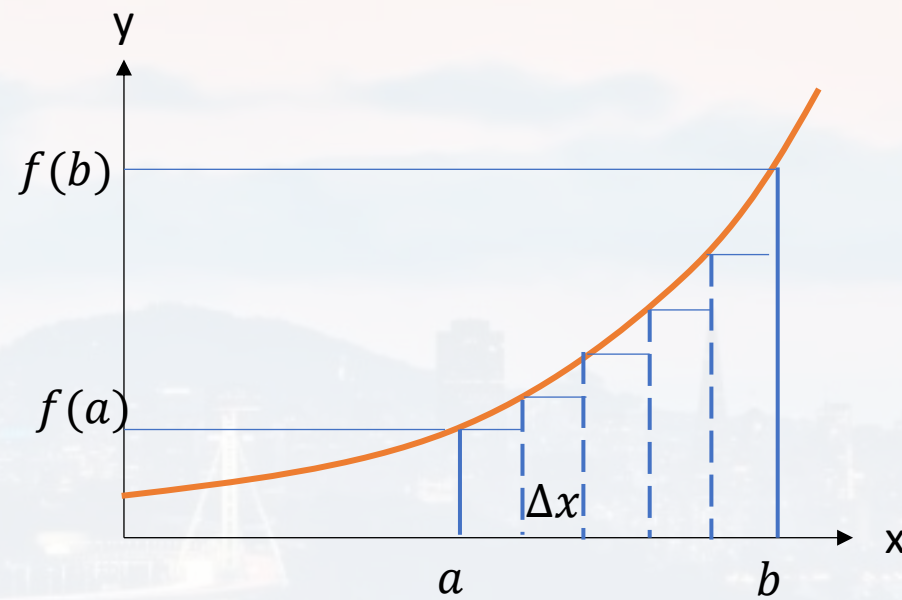


area under a curve (1D)

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

$$N = \frac{b - a}{\Delta x}$$

$$A_{tot} \approx$$



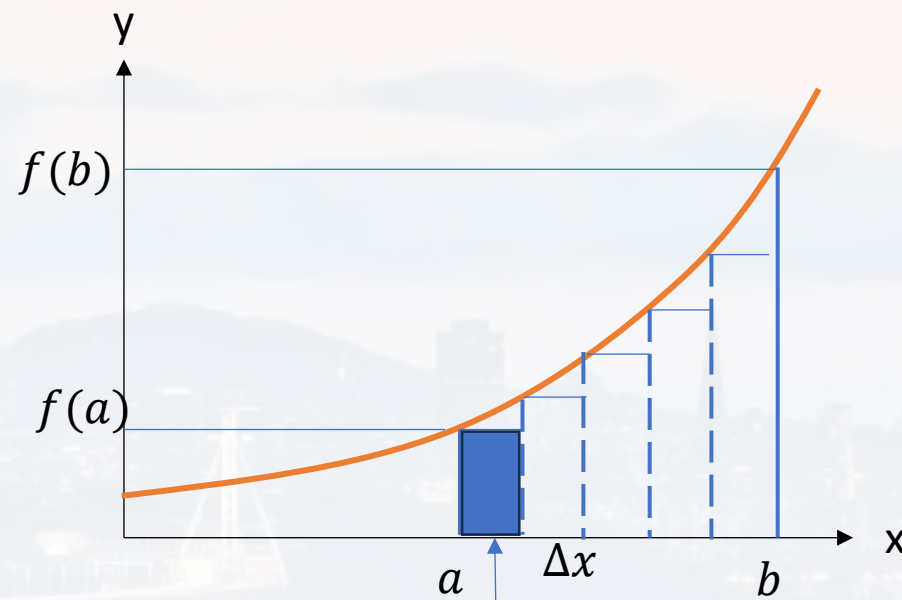


area under a curve (1D)

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

$$N = \frac{b - a}{\Delta x}$$

$$A_{tot} \approx f(a) \Delta x$$



$$A_0 = f(a) \Delta x$$

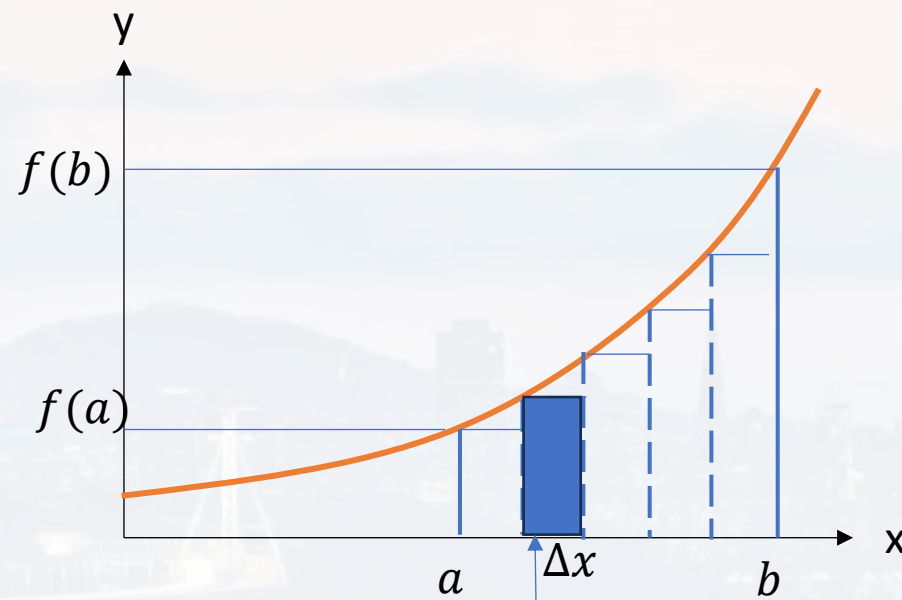


area under a curve (1D)

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

$$N = \frac{b - a}{\Delta x}$$

$$A_{tot} \approx f(a) \Delta x + f(a + \Delta x) \Delta x$$



$$A_1 = f(a + \Delta x) \Delta x$$

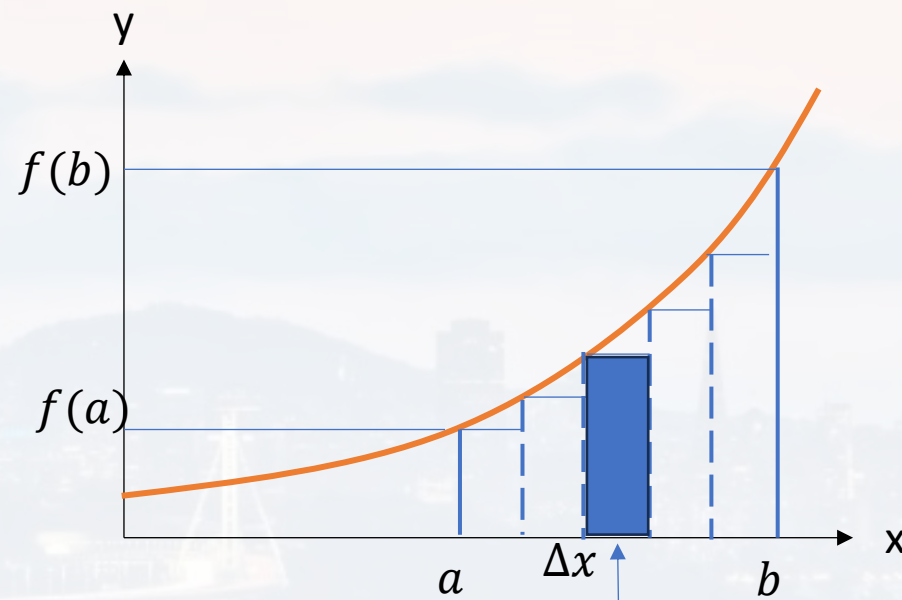


area under a curve (1D)

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

$$N = \frac{b - a}{\Delta x}$$

$$A_{tot} \approx f(a) \Delta x + f(a + \Delta x) \Delta x + f(a + 2\Delta x) \Delta x$$



$$A_2 = f(a + 2\Delta x) \Delta x$$

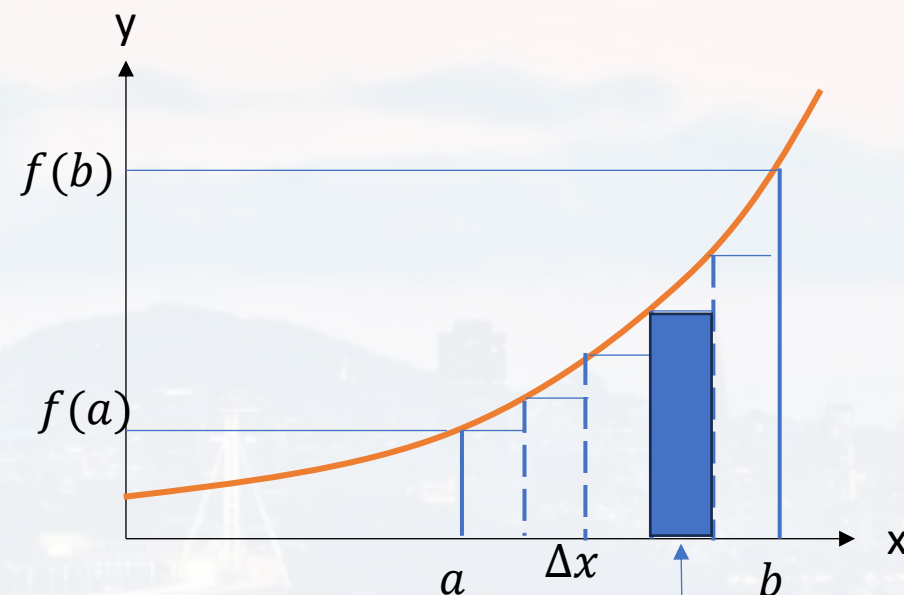


area under a curve (1D)

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

$$N = \frac{b - a}{\Delta x}$$

$$A_{tot} \approx f(a) \Delta x + f(a + \Delta x) \Delta x + f(a + 2\Delta x) \Delta x + f(a + 3\Delta x) \Delta x$$



$$A_3 = f(a + 3\Delta x) \Delta x$$



area under a curve (1D)

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

$$N = \frac{b - a}{\Delta x}$$

$$A_{tot} \approx f(a + \mathbf{0}\Delta x) \Delta x$$

$$+ f(a + \mathbf{1}\Delta x) \Delta x$$

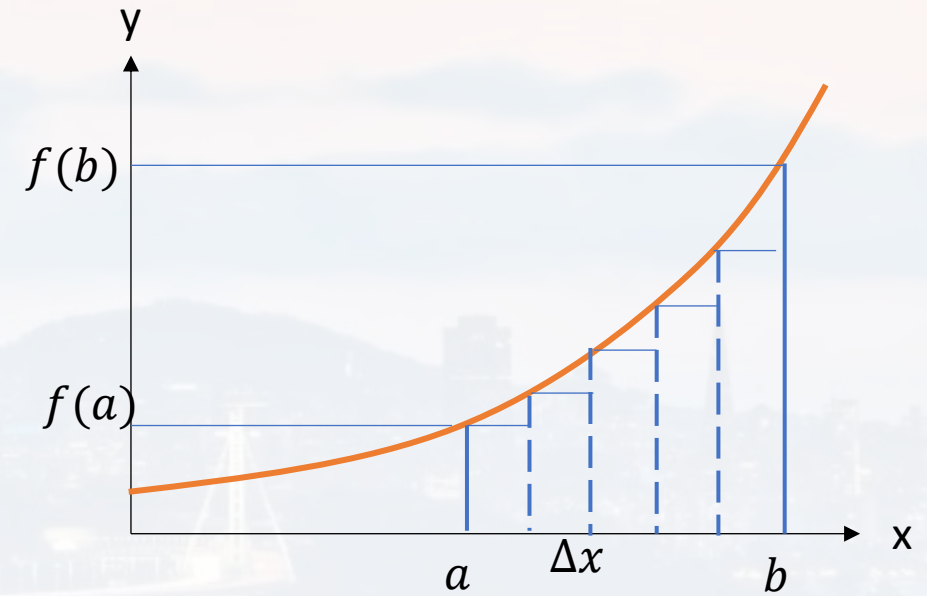
$$+ f(a + \mathbf{2}\Delta x) \Delta x$$

$$+ f(a + \mathbf{3}\Delta x) \Delta x$$

$$+ f(a + \mathbf{4}\Delta x) \Delta x$$

$$+ \dots$$

$$= \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$





area under a curve (1D)

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

$$N = \frac{b - a}{\Delta x}$$

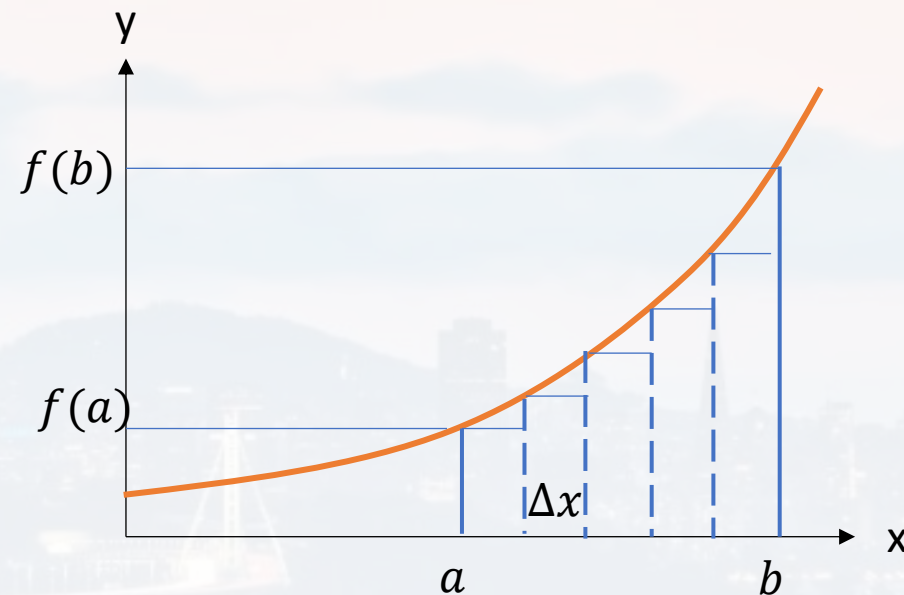
more accurate:

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [f(a + i \Delta x) + f(a + (i + 1) \Delta x)] \frac{\Delta x}{2}$$

trapezoidal rule

error (for large N):

$$\varepsilon = -\frac{(b - a)^2}{12 N^2} [f'(b) - f'(a)] + O(N^{-3})$$





$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [f(a + i \Delta x) + f(a + (i+1) \Delta x)] \frac{\Delta x}{2} \quad N = \frac{b-a}{\Delta x}$$

example: $f(x) = x^2$

$$\int_a^b x^2 dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [(a + i \Delta x)^2 + (a + (i+1) \Delta x)^2] \frac{\Delta x}{2}$$

$$= \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [2a^2 + i^2 \Delta x^2 + 2ai \Delta x + a^2 + (i+1)^2 \Delta x^2 + 2a(i+1) \Delta x] \frac{\Delta x}{2}$$

$$= \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [2a^2 + i^2 \Delta x^2 + 2ai \Delta x + i^2 \Delta x^2 + \Delta x^2 + 2i \Delta x^2 + 2ai \Delta x + 2a \Delta x] \frac{\Delta x}{2}$$



example:

$$f(x) = x^2$$

$$N = \frac{b - a}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [2a^2 + i^2 \Delta x^2 + 2ai\Delta x + i^2 \Delta x^2 + \Delta x^2 + 2i\Delta x^2 + 2ai\Delta x + 2a\Delta x] \frac{\Delta x}{2}$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 N + \Delta x^3 \sum_{i=0}^{N-1} i^2 + 2a\Delta x^2 \sum_{i=0}^{N-1} i + \frac{\Delta x^3}{2} N + \Delta x^3 \sum_{i=0}^{N-1} i + a\Delta x^2 N \right]$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 N + \Delta x^3 \sum_{i=0}^{N-1} i^2 + 2a\Delta x^2 \sum_{i=0}^{N-1} i + \Delta x^3 \sum_{i=0}^{N-1} i \right]$$



example: $f(x) = x^2$

$$N = \frac{b - a}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 N + \Delta x^3 \sum_{i=0}^{N-1} i^2 + 2a\Delta x^2 \sum_{i=0}^{N-1} i + \Delta x^3 \sum_{i=0}^{N-1} i \right]$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 N + \Delta x^3 \frac{N(N+1)(2N+1)}{6} + a\Delta x^2 N(N+1) + \Delta x^3 \frac{N(N+1)}{2} \right]$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 N + \Delta x^3 \frac{N(N+1)(2N+1)}{6} + a\Delta x^2 N(N+1) \right]$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 N + \Delta x^3 \frac{(N^2 + N)(2N+1)}{6} + a\Delta x^2 N^2 + a\Delta x^2 N \right]$$



example: $f(x) = x^2$

$$N = \frac{b - a}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 N + \Delta x^3 \frac{(N^2 + N)(2N + 1)}{6} + a \Delta x^2 N^2 + a \Delta x^2 N \right]$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 N + \Delta x^3 \frac{2N^3 + 3N^2 + N}{6} + a \Delta x^2 N^2 \right]$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 \frac{b - a}{\Delta x} + \Delta x^3 \frac{2 \left(\frac{b - a}{\Delta x} \right)^3 + 3 \left(\frac{b - a}{\Delta x} \right)^2 + \frac{b - a}{\Delta x}}{6} + a \Delta x^2 \left(\frac{b - a}{\Delta x} \right)^2 \right]$$

$$= a^2(b - a) + \frac{(b - a)^3}{3} + a(b - a)^2 = \frac{b^3 - a^3}{3}$$

$$\int_a^b x^2 dx = \frac{b^3 - a^3}{3}$$



$$a, c \in \mathbb{C}$$
$$n \in \mathbb{R}$$

rules: $\int \frac{d}{dx} f(x) dx = \int df(x) = f(x) + c$

therefore: an integral is
an **anti derivative!**

$$\int ax^n dx = a \frac{1}{n+1} x^{n+1} + c \quad n \neq -1$$

$$\int f(x) + g(x) dx = \int f(x) dx + \int g(x) dx + c$$

sum rule: integrals are linear

$$\int f(x) \frac{d}{dx} g(x) dx = f(x)g(x) + \int \frac{d}{dx} f(x) \cdot g(x) dx + c$$

product rule



special integrals

$$\int e^x dx = e^x + c$$

$$\int a^x dx = \frac{a^x}{\ln(a)} + c$$

$$\int \frac{1}{x} dx = \ln(|x|) + c$$

$$\int \log_b(x) dx = x \log_b(x) - \frac{x}{\ln(b)} + c$$

$$\int \cos(x) dx = \sin(x) + c$$

$$\int \sin(x) dx = -\cos(x) + c$$

$$a, c \in \mathbb{C}$$
$$n \in \mathbb{R}$$



$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

$$N = \frac{b - a}{\Delta x}$$

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [f(a + i \Delta x) + f(a + (i + 1) \Delta x)] \frac{\Delta x}{2}$$

trapezoidal rule

even more accurate:

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [f(a + i \Delta x) + f(a + (i + 1) \Delta x) + 4 f(a + i \Delta x/2)] \frac{\Delta x}{6}$$

Simpson rule

Note: there are different Simpson rules, depending on how many subintervals are included



Newton-Cotes Equations

approximation

error

$$\frac{1}{2} \Delta x (f_i + f_{i+1})$$

$$\varepsilon \sim \frac{\Delta x^3}{12}$$

trapezoidal

$$\frac{1}{6} \Delta x (f_i + f_{i+2} + 4f_{i+1})$$

$$\varepsilon \sim \frac{\Delta x^5}{90}$$

Simpson

$$\frac{1}{8} \Delta x (f_i + 3f_{i+1} + 3f_{i+2} + f_{i+3})$$

$$\varepsilon \sim \frac{3 \Delta x^5}{80}$$

Simpson 3/8

$$\frac{1}{90} \Delta x (7f_i + 32f_{i+1} + 12f_{i+2} + 32f_{i+3} + 7f_{i+4})$$

$$\varepsilon \sim \frac{8 \Delta x^7}{945}$$

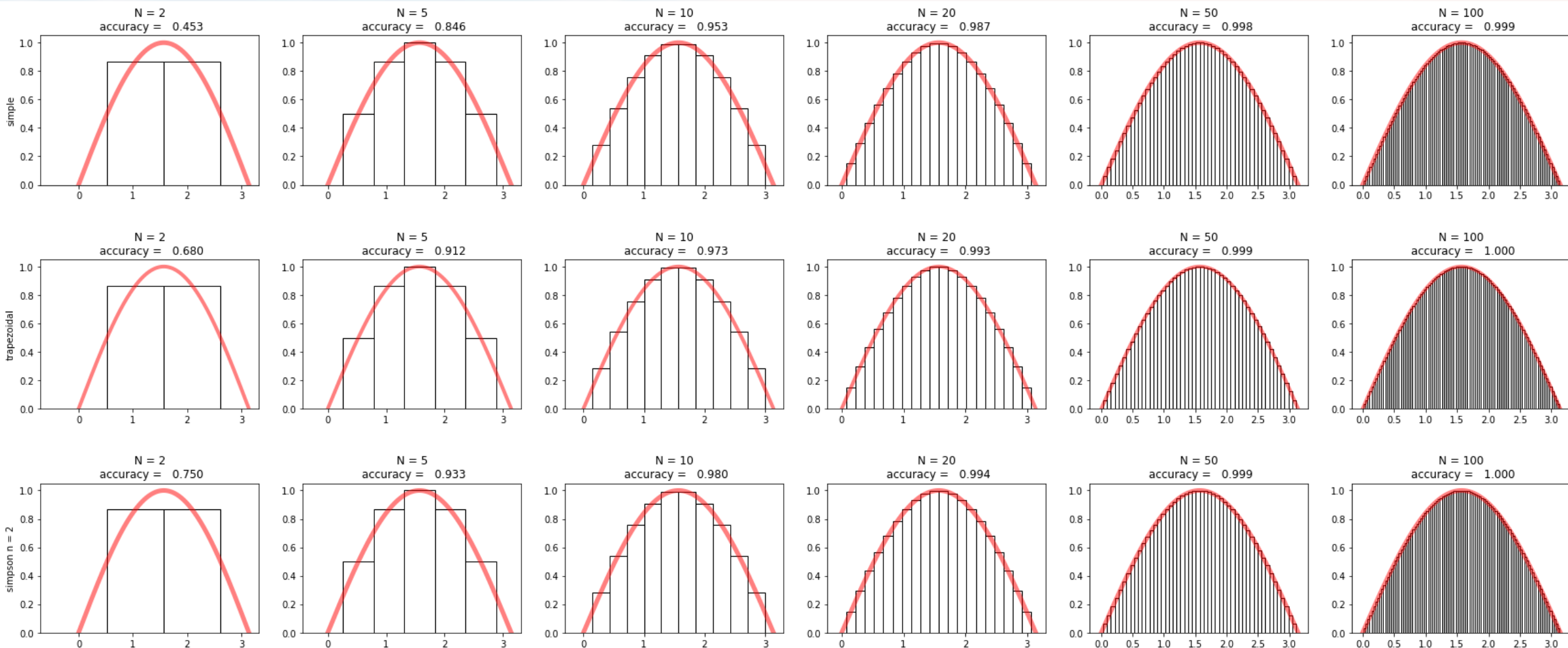
Boole

Note: i here refers to subinterval **within** Δx



run the function `IntegrationAccuracy.py`

integrating $\sin(x)$





SciPy

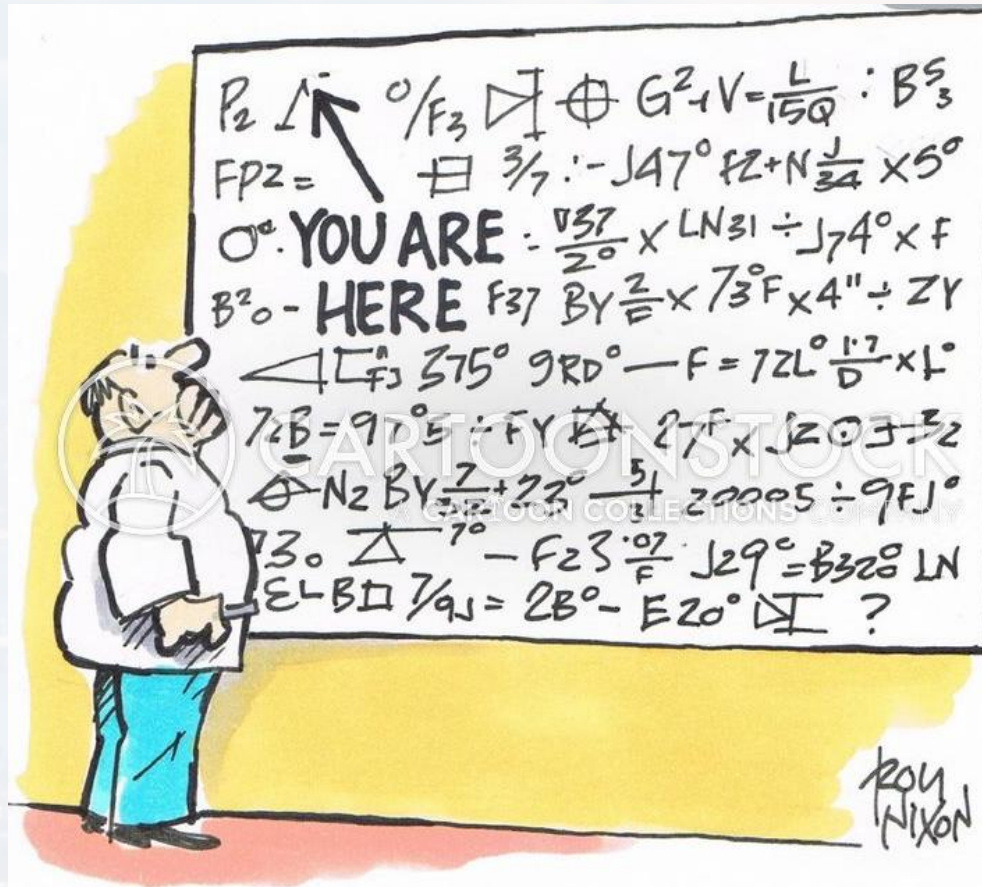
```
from scipy.integrate import *
```

```
simpson  
trapezoid  
quad
```

...and more.

See homework assignment!

```
I = simpson(y, x)
```



Outline:

Derivatives

Taylor Series

Integrals

Diffusion Example



often, we need to **model diffusion processes** (heat conduction, diffusion reaction, electromagnetism etc)

$$\frac{\partial c(x, y, z, t)}{\partial t} = D \Delta c(x, y, z, t)$$

c :

concentration

D :

diffusion constant

$$\Delta \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} :$$

Laplace operator

numerically:

$$\frac{c(x_0, y_0, t_0 + \Delta t) - c(x_0, y_0, t_0 - \Delta t)}{2\Delta t} = D \left[\right.$$

$$\frac{c(x_0 + \Delta x, y_0, t_0) - 2c(x_0, y_0, t_0) + c(x_0 - \Delta x, y_0, t_0)}{\Delta x^2} +$$

$$\left. \frac{c(x_0, y_0 + \Delta y, t_0) - 2c(x_0, y_0, t_0) + c(x_0, y_0 + \Delta y, t_0)}{\Delta y^2} \right]$$



often, we need to **model diffusion processes** (heat conduction, diffusion reaction, electromagnetism etc)

$$\frac{\partial c(x, y, z, t)}{\partial t} = D \Delta c(x, y, z, t)$$

c : concentration
 D : diffusion constant
 $\Delta \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$: Laplace operator

numerically:

$$c(x_0, y_0, t_0 + \Delta t) =$$

We can calculate c
in the **future**

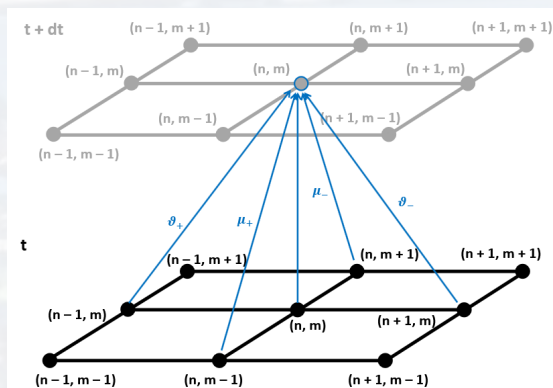
$$2\Delta t D \left[\frac{c(x_0 + \Delta x, y_0, t_0) - 2c(x_0, y_0, t_0) + c(x_0 - \Delta x, y_0, t_0)}{\Delta x^2} + \right.$$

by using all adjacent
current values

$$\left. \frac{c(x_0, y_0 + \Delta y, t_0) - 2c(x_0, y_0, t_0) + c(x_0, y_0 + \Delta y, t_0)}{\Delta y^2} \right]$$

$$+ c(x_0, y_0, t_0 - \Delta t)$$

and the immediate
past value



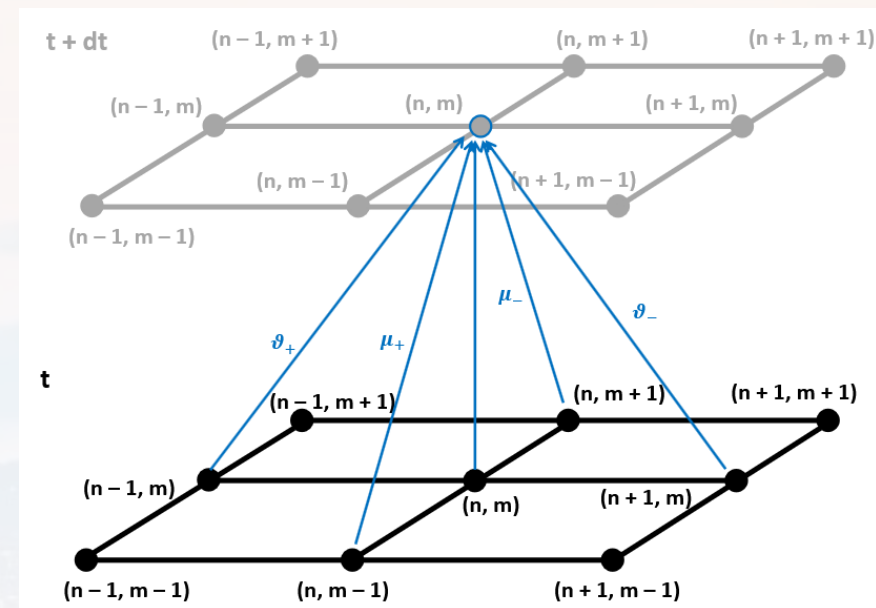
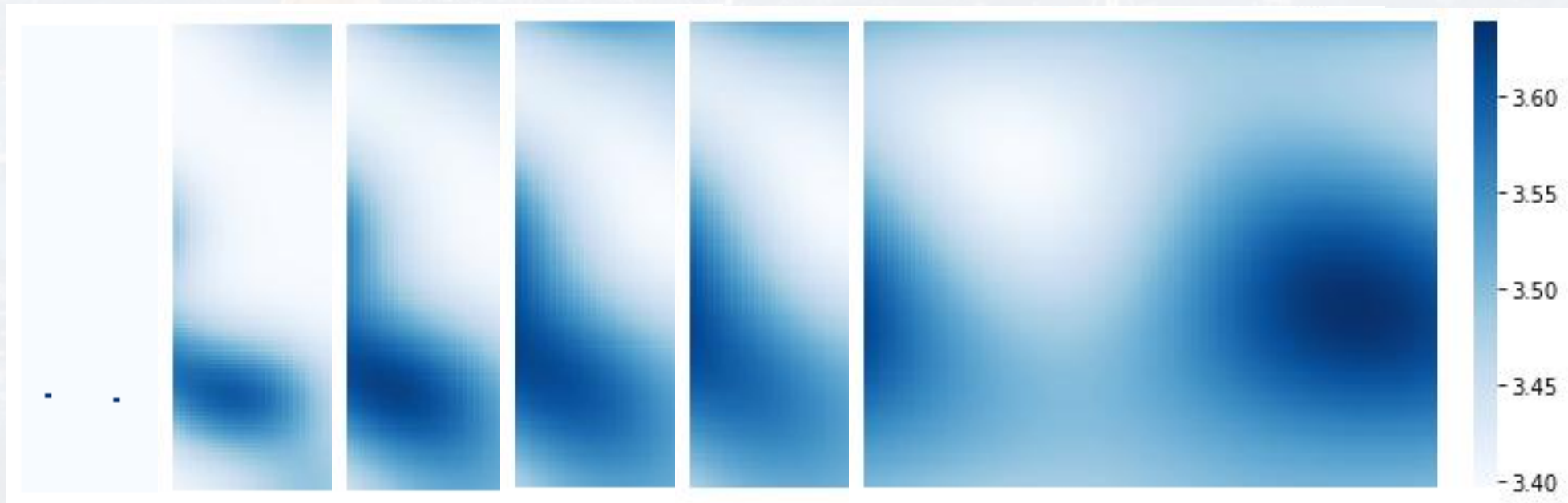


run the script Diffusion2D.py

```
from Diffusion2D import *
```

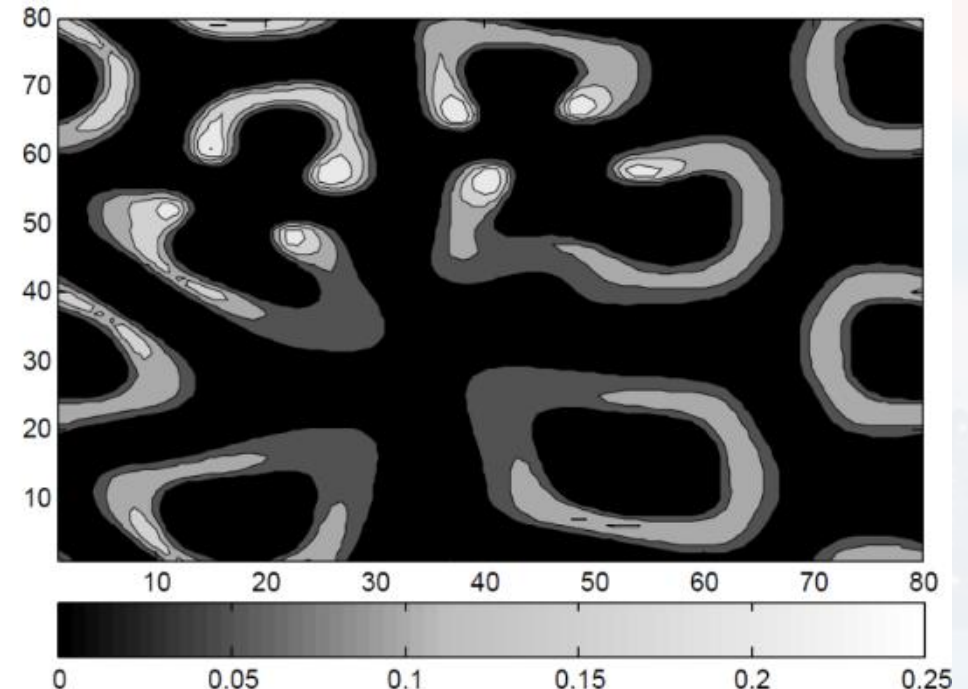
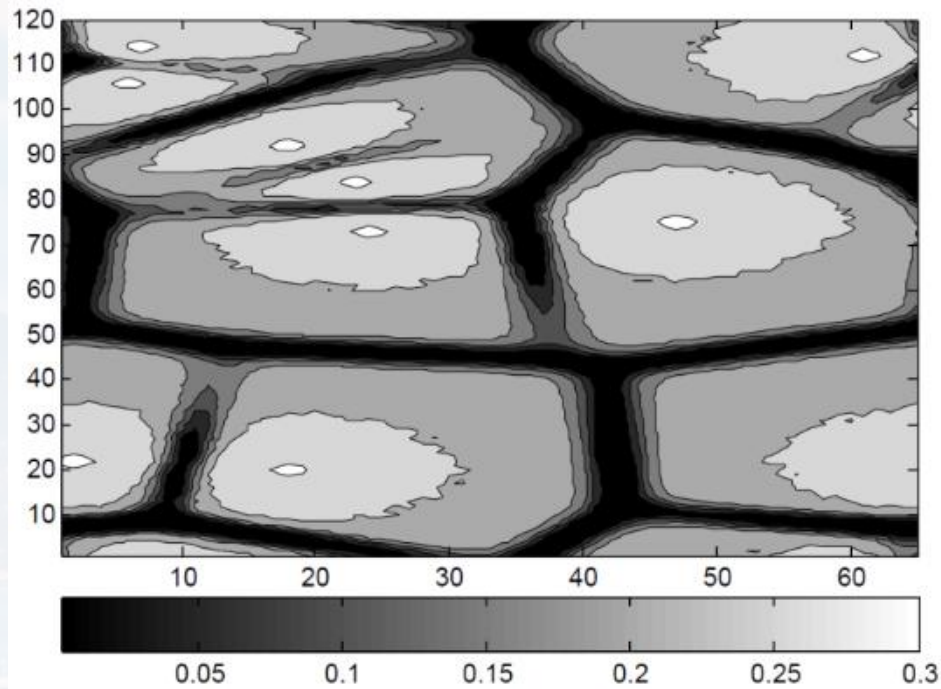
```
D = Diffusion2D()  
D.RunSimulation()
```

initial condition





Possible Capstone Project: modelling fur and skin pattern:





Thank you very much for your attention!

